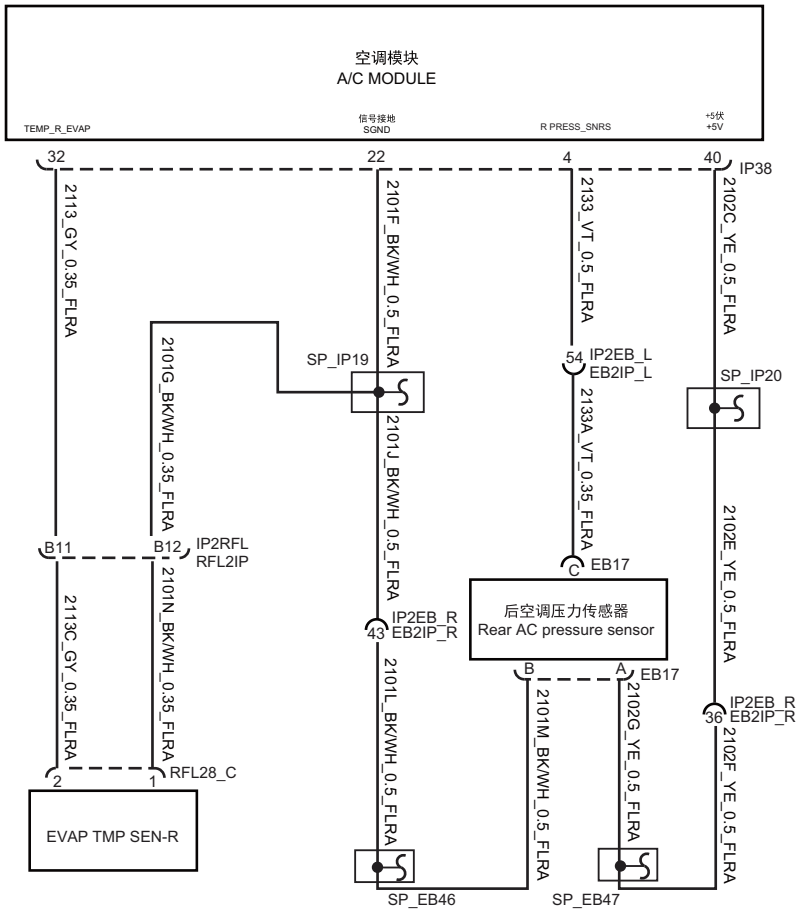
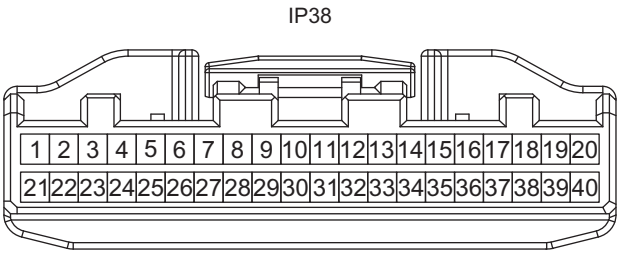
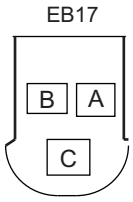
## U0197-87

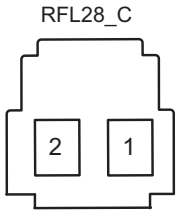
| Description of fault diagnosis codes |
| --- |
| U0197-87：R Press  sensor short or open |
| **Detection tools** |
| 1 Multimeter and diagnostic apparatus |
| **Possible causes** |
| • Circuit,terminal and adapting piece fault<br>• Air conditioning controller fault<br>• R Press sensor of air conditioning fault |
| **Conditions for fault code operation** |
| The ignition switch is in ON position. |
| **Conditions for fault code setting** |
| U0197-87 |
| |
| **To set the effect of a fault code condition** |
| 1<br>2 |
| **Description of circuit diagram** |
| The air conditioning control module monitors the R press sensor via low level reference circuit and signal circuit. |

**Circuit diagram**

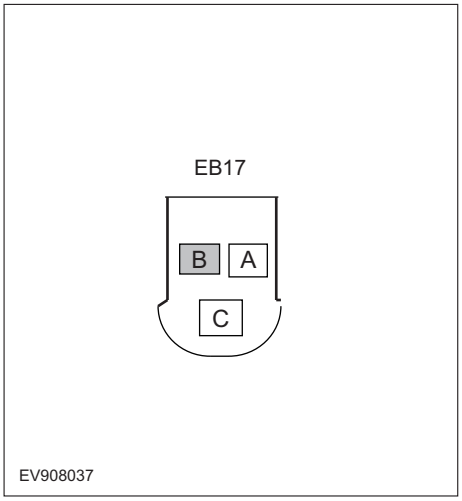


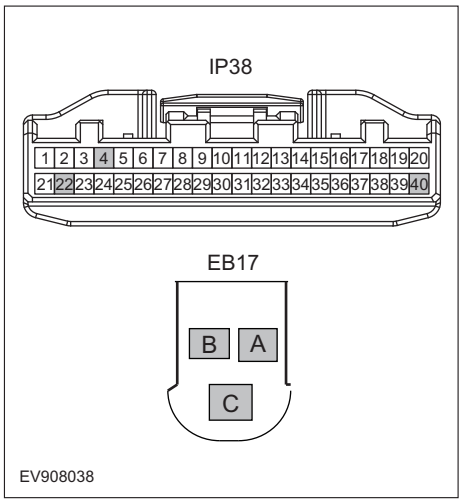




EV908036

| Troubleshooting procedures | | |
|---|---|---|
| Step 1 | General inspection. | |
| | | 1 Check the fuse for damage.<br>2 Check battery capacity.<br>3 Check the harness plug of R press sensor of air conditioning and air conditioning control for damage, poor contact, aging and looseness.<br>4 Check whether the result is normal or not? |
| | | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | | |
| Step 2 | Check fault codes. | |
| | | 1 Connect the fault diagnosis instrument to diagnosis interface.<br>2 Turn the ignition switch to ON position.<br>3 Diagnose the air condition control system with diagnostic apparatus.<br>4 Is there any other fault code except for U0197-87? |
| | | **YES →** Refer to:DTC Summary List (HVAC). |
| **NO ↓** | | |
| Step 3 | Check the power supply circuit of R press sensor of air conditioning. | |
|  | | 1 Turn the ignition switch to OFF.<br>2 Disconnect the harness plug EB17 of R press sensor of air conditioning.<br>3 Turn the ignition switch to ON.<br>4 Check the voltage between the plug terminal of R press sensor of air conditioning and ground.<br>Measuring circuit: Voltage between terminal B on plug EB17 and ground.<br>**Standard value:4.8-5.2V**<br>5 Check whether the result is normal or not? |
| | | **NO →** Please repair or replace the faulted parts. |
| **YES ↓** | | |

| Step 4 | Check the circuit of R press sensor of air conditioning (open circuit). |
|---|---|




1 Turn the ignition switch to OFF.

2 Disconnect the harness plug IP38 of air conditioning controller.

3 Check the resistance between harness plugs of R press sensor of air conditioning and of air conditioning controller.

Measuring circuit: resistance between terminal A on plug EB17 and terminal 22 on plug IP38.

Measuring circuit: resistance between terminal B on plug EB17 and terminal 40 on plug IP38.

Measuring circuit: resistance between terminal C on plug EB17 and terminal 4 on plug IP38.

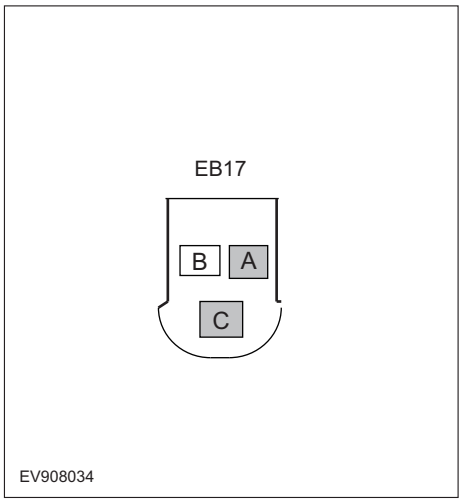
**Standard value: ≈ 0Ω**

4 Check whether the result is normal or not?

**NO →** Please repair or replace the faulted parts.

**YES ↓**

| Step 5 | Check the circuit of R press sensor of air conditioning (short circuit). |
|---|---|




1 Turn the ignition switch to OFF.

2 Check the resistance between the plug terminal of R press sensor of air conditioning and ground.

Measuring circuit: resistance between terminal A on plug EB17 and ground.

Measuring circuit: resistance between terminal C on plug EB17 and ground.

**Standard value: ∞**

3 Check the voltage between the harness plug of R press sensor of air conditioning and ground.

Measuring circuit: voltage between terminal A on plug EB17 and ground.

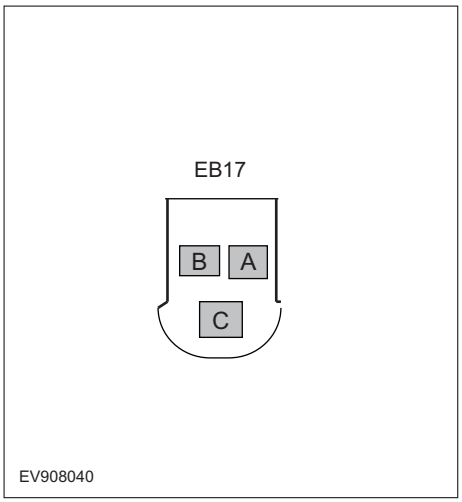
Measuring circuit: voltage between terminal C on plug EB17 and ground.

**Standard value: ≈ 0V**

4 Check whether the result is normal or not?

**NO →** Repair or replace the faulted parts.

**YES ↓**

| Step 6 | Check the R press sensor of air conditioning (parts). |
|---|---|
|  | 1 Measure the resistance of terminals on R press sensor of air conditioning directly.<br>Measuring circuit: measure the resistance between terminal A and B on plug EB17.<br>**Standard value:**<br>2 Measure the resistance between the R press sensor terminal of air conditioning and the housing.<br>Measuring circuit: Measure the resistance between the terminal A on EB17 and the housing.<br>Measuring circuit: Measure the resistance between the terminal B on EB17 and the housing.<br>Measuring circuit: Measure the resistance between the terminal C on EB17 and the housing.<br>**Standard value: ∞**<br>3 Check whether the result is normal or not? |
| | **NO →** Replace the R press sensor of air conditioning. |
| **YES ↓** | |
| Step 7 | Check if air conditioning module is normal. |
| | 1 Turn the ignition switch to OFF.<br>2 Disconnect and check the harness plug IP38 of air conditioning controller.<br>3 Check and repair;<br>• Corrosion(install new adapting piece or terminal-cleaning module and pin).<br>• Pin damaged or bent-install new terminal/pin.<br>• Fly out of pins-install new pins as needed.<br>4 Reconnect the air conditioning control module plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed.<br>5 Run the system and determine if the problem persists? |
| | **YES →** Replace the air conditioning control module. |
| **NO ↓** | |
| At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems, | |
| **Next Step ↓** | |
| Diagnosis end. | |